Technical Supplement to Decaying Autoregressive Memory Models for Conditional Volatility:

Mathematical Foundations, Finite-Sample Identification and Out-of-Sample Evidence

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DARM Preprint v1.0
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Supplementary proofs, expanded results and reproducibility notes

Abstract

This independently compilable supplement accompanies the revised DARM preprint. It retains full reported tables, secondary simulation analyses and audit qualifications from Experiments 13–18, together with supporting derivations. The empirical figures are reproduced from supplied historical or simulation outputs; no new experiment was performed. Original source implementations for Experiments 13–18 are now collected in a separate research archive, but the underlying return/latent-state paths and the full equity price panel remain incomplete.

1 Scope and relation to the main paper

All claims about statistical significance, model comparison and forecast timing are governed by the explicit limitations of the main paper. In particular, the historical ten-day primary comparison remains inconclusive; these synthetic results do not constitute new financial-market evidence. The sections are organised by experiment to facilitate traceability rather than to define the main preprint's conceptual structure.

2 Additional independent-market diagnostics

The archived four-, eight- and twelve-week bootstrap intervals for the primary asymmetric-DARM ten-day gain were

$[-0.006829, 0.032128]$, $[-0.008774, 0.031254]$ and $[-0.008289, 0.030559]$, respectively. The corresponding bootstrap p-values were 0.126437, 0.123938 and 0.119940. All include zero. Exploratory sector-neutral portfolio risk-weighting reduced observed variance but did not establish superior DARM economic performance; portfolio-level 10-day QLIKE improvement was 0.000655 with reported HAC $t = 0.0633$. These results do not replace the primary test.

3 Mathematical companion details

The next statements extend the main paper's formulas without new statistical claims. For a finite-K state $\boldsymbol{S}_t = (u_t, u_{t-1}, \ldots, u_{t-K+1})^\top$ and $\boldsymbol{M}_t$ with top row $A_t\rho(w_1, \ldots, w_K)$ and unit subdiagonal, define $Q = \mathbb{E}(\boldsymbol{M}_t \otimes \boldsymbol{M}_t)$. If $\mathbb{E}A_t^2 < \infty$, $\rho\mathbb{E}A_t < 1$ and $\mathrm{spr}(Q) < 1$, the nonnegative Neumann series

gives a finite state second moment, using finite first moments for cross terms. This is a sufficient criterion and is not stated as a universal equivalence. For $K = 1$, the exact criterion with positive intercept is $\rho^2 \mathbb{E}A^2 < 1$ and

$$\mathbb{E}h_t^2 = \frac{\omega^2(1+\rho m)}{(1-\rho m)(1-\rho^2\mathbb{E}A^2)}, \qquad m = \mathbb{E}A.$$

4 Corrected centred-variance and spectral filters

This section records the correction in Theorem 5.1 of the accompanying main paper, independently of the shorthand $m = 1$ used for symmetric innovation laws. Let $m = \mathbb{E}A_t \in (0, \infty)$ and assume a causal stationary solution with $\mathbb{E}u_t^2 < \infty$, $\rho m < 1$. Define

$$\zeta_t = u_t - mh_t, \quad Y_t = u_t - \mathbb{E}u_t, \quad \chi_t = \varepsilon_t^2 - h_t, \quad c_k = \rho m w_k.$$

Here ζ_t is a square-integrable martingale difference; χ_t is square-integrable if $\mathbb{E}\varepsilon_t^4 < \infty$. Writing $\Psi_m(s) = (1 - \rho m W(s))^{-1} = \sum_{j\geq 0} \psi_j s^j$, the renewal equation gives $Y_t = \sum_{j\geq 0} \psi_j \zeta_{t-j}$ with $\psi_0 = 1$. Since $Y_t = \zeta_t + m(h_t - \mathbb{E}h_t)$,

$$\boxed{h_t - \mathbb{E}h_t = \frac{Y_t - \zeta_t}{m} = \frac{1}{m}\sum_{j\geq 1} \psi_j \zeta_{t-j}} \tag{1}$$

with no second contemporaneous subtraction. Define the strictly causal filter

$$D_m(s) = \frac{\Psi_m(s) - 1}{m} = \frac{\rho W(s)}{1 - \rho m W(s)} = \sum_{j\geq 1} \frac{\psi_j}{m} s^j.$$

Thus $\varepsilon_t^2 - \mathbb{E}\varepsilon_t^2 = \chi_t + D_m(L)\zeta_t$ and, with finite second moments of both martingale differences, their contemporaneous covariance is

$$\sigma_{\chi\zeta} = \mathbb{E}h_t^2\{\mathbb{E}[z_t^2 A_t] - m\}.$$

Orthogonality across distinct dates (not independence within a date) gives

$$f_h(\lambda) = \frac{\sigma_\zeta^2}{2\pi}|D_m(e^{-i\lambda})|^2,$$
$$f_{\varepsilon^2}(\lambda) = \frac{1}{2\pi}\left[\sigma_\chi^2 + \sigma_\zeta^2|D_m(e^{-i\lambda})|^2 + 2\sigma_{\chi\zeta}\Re D_m(e^{-i\lambda})\right].$$

Under symmetric standardised innovations, $m = 1$; the main-paper $D = C/(1 - C)$ and cross-spectrum formulas follow exactly. Since $\sum_{j\geq 1} \psi_j/m < \infty$ and χ, ζ are orthogonal through time, the squared-return covariance remains absolutely summable. This repairs the *proof identity*, not the claimed summability or the final spectral expressions. The filtered martingale differences may be contemporaneously correlated under asymmetry.

5 Complete Monte Carlo tables and protocol notes

6 Synthetic identification and finite-sample likelihood inference

The following results concern three connected *synthetic* Monte Carlo exercises. They do not extend the preceding historical backtest, alter its test statistics, or establish new investor returns. Under the fixed generating law, $K = 250$, $\rho_0 = 0.75$, $\alpha_0 = 1.8$, $\delta_0 = 5$, $\nu_0 = 8$, and standardised Student-t innovations, Experiments 13–15 examine finite-sample parameter recovery and profile-likelihood coverage. The two first studies use both symmetric ($\kappa = 0$) and asymmetric ($\kappa = 0.6$) processes; the third investigates $\kappa_0 \in \{0.2, 0.6, 0.9\}$.

6.1 Scope, sample construction and reproducibility

All synthetic generators discard a 2,000-step burn-in. For estimation, the first K observed returns are conditioned upon and omitted from the Student-t likelihood, so the effective number of contributions is $T - K$; the nominal T denotes simulated retained training observations. Estimation uses multi-start bounded L-BFGS-B optimisation. The numerical admissible box includes $\alpha \in [10^{-4}, 10]$, $\delta \in [10^{-4}, 100]$ and, for asymmetric fits, $\kappa \in [0, 0.999]$. These are computational restrictions, not parameter-support boundaries of an unconstrained power-law family.

The consolidated reported-results CSV contains 775 rows of *reported aggregate metrics* spanning a theory-validation pilot (Experiment 12) and full Experiments 13–15. The provided Experiment 15 ZIP comprises four executed notebooks but no observation-level Monte Carlo outputs, likelihood-fit checkpoint JSONs or fitted parameter vectors; hence the tabulated quantities are verified against the consolidated summaries and notebook-generated outputs, not independently recomputed from the individual simulated sample paths. Re-executing all optimisations would constitute a separate computational replication. Experiment 12 is not pooled with the full Monte Carlo estimates.

A relevant dependence across studies is observable directly in source code: Experiments 13 and 14 share the same master seed, sample-seed formula, generating parameters and equivalent sample generators. Their matched configuration/replication index therefore reconstructs the same training returns (up to deterministic implementation equivalence). These are *paired* investigations of recovery and inference, not 600 independent synthetic training datasets. Experiment 15 uses a different configuration-dependent seed formula.

6.2 Experiment 13: parameter recovery and oracle-relative forecasting

There are 50 fitted replications at each of six $(\text{symmetry}, T)$ combinations, 300 fits in total. Table 1 reproduces the principal aggregate summaries. In all cells estimated α and δ are much noisier than the persistence parameter ρ. Larger T generally reduces error, though the rates are not established by six aggregate points. The "variance correlation" is between fitted and *true* out-of-sample conditional-variance paths in a 750-period synthetic holdout, not correlation with realised squared returns.

Table 1: Experiment 13: reported 50-replication means or RMSEs. $\overline{D}_Q$ is mean conditional *expected* QLIKE regret relative to the true conditional variance; not realised-data QLIKE. "Bound" is the share of fits with at least one coefficient within the notebook's boundary tolerance.

Model	T	RMSE α	RMSE δ	Corr$(\hat{h}, h_0)$	$\overline{D}_Q$	Bound (%)
Symmetric	1,500	3.694	27.149	0.98446	0.00564	18
Symmetric	4,000	1.836	16.218	0.99408	0.00199	4
Symmetric	8,000	0.446	2.956	0.99803	0.00075	0
Asymmetric	1,500	4.272	28.630	0.97570	0.00747	36
Asymmetric	4,000	0.552	3.484	0.99239	0.00199	0
Asymmetric	8,000	0.283	1.989	0.99722	0.00101	0

The score used for expected regret is especially informative. Writing $r_t = \hat{h}_t / h_{0,t} > 0$, the study computes

$$D_Q(r_t) = \log r_t + \frac{1}{r_t} - 1 \geq 0, \qquad D_Q(1+e) = \tfrac{1}{2}e^2 + O(e^3), \tag{2}$$

which follows by taking the conditional expectation of the ordinary QLIKE score difference under the true model. Thus low reported regret is a *synthetic oracle* diagnostic that is unavailable with unknown real-world volatility. The Spearman correlations between $|\hat{\alpha} - \alpha_0|$, $|\hat{\delta} - \delta_0|$ or kernel discrepancy

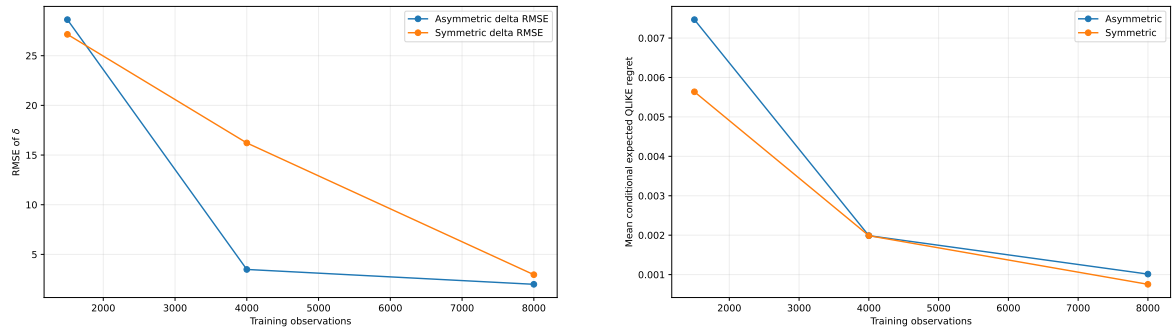


Figure 1: Experiment 13 reported aggregates: parameter-recovery error (left) and oracle-relative expected QLIKE regret (right). Each plotted point summarises 50 Monte Carlo replications; curves are descriptive, not estimated convergence rates.

and regret are exploratory across-replication associations (24 unadjusted tests), not independent confirmatory evidence.

6.3 Experiment 14: primary profile inference for δ

Fifty replications in each of the same six configurations yield 300 primary δ LR tests. At each candidate δ, all nuisance coefficients are reoptimised. For an effective contribution count $n = T - K$, the profile statistic is

$$LR_n(\delta_0) = 2\{\ell_n(\widehat{\Theta}) - \sup_{\Theta:\delta=\delta_0} \ell_n(\Theta)\}, \qquad \text{nominal 95\% acceptance when } LR_n \le \chi^2_{1,0.95}. \tag{3}$$

The study reports coverage directly at the generating value. It also computes the locally connected interval around the estimated maximum; where a likelihood region is disconnected, the displayed endpoints need not describe the full confidence set. Table 2 demonstrates how nominal coverage can coincide with very weak precision. Secondary coverage summaries for α and ρ (and κ in the

Table 2: Experiment 14: primary δ profiles, 50 replications per cell. The width is the *median* of reported local intervals. The upper-bound hit share is based on the numerical limit $\delta = 100$.

Model	T	Coverage (%)	Median width	Upper bound (%)
Symmetric	1,500	96	95.156	48
Symmetric	4,000	94	48.606	8
Symmetric	8,000	96	11.606	0
Asymmetric	1,500	94	78.952	32
Asymmetric	4,000	96	16.807	0
Asymmetric	8,000	94	9.630	0

asymmetric model) use only *12 profiles per configuration*, not 50. Their Wilson intervals are extremely broad, so they are reported as exploratory diagnostics rather than calibrated confidence statements.

6.4 Experiment 15: inference for κ

For $\kappa_0 \in \{0.2, 0.6, 0.9\}$ and $T \in \{1500, 4000\}$, the code runs 100 unconditional synthetic replications per cell. It constructs the likelihood ratio at the *true* κ_0 by reoptimising the remaining five coefficients at the null value, after multi-start unrestricted estimation. Table 3 displays the six reported 95% acceptance rates, the associated Wilson intervals for the unknown coverage probability, and fitted-κ

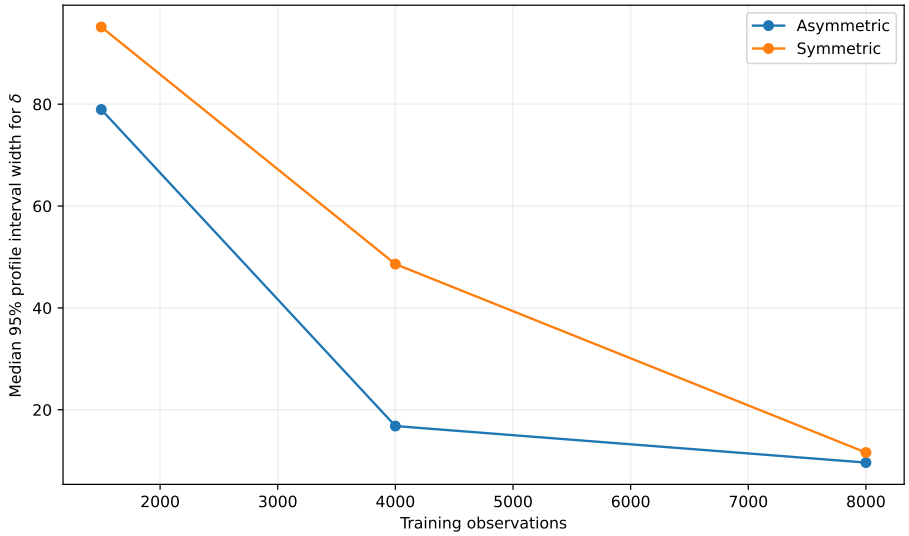


Figure 2: Experiment 14: shrinking median δ interval widths. The wide small-sample sets and numerical bound hits are evidence of weak finite-sample precision despite approximately nominal coverage.

Table 3: Experiment 15: 600 primary LR tests; 100 valid fits per cell, no reported failures. "Upper MLE" uses the $\hat{\kappa} \geq 0.998$ threshold. Wilson bounds quantify Monte Carlo uncertainty in observed coverage.

κ_0	T	Covered / 100	Wilson lower	Wilson upper	Upper MLE (%)
0.2	1,500	95	0.8882	0.9785	0
0.2	4,000	94	0.8752	0.9722	0
0.6	1,500	93	0.8625	0.9657	7
0.6	4,000	90	0.8256	0.9448	0
0.9	1,500	100	0.9630	1.0000	37
0.9	4,000	97	0.9155	0.9897	20

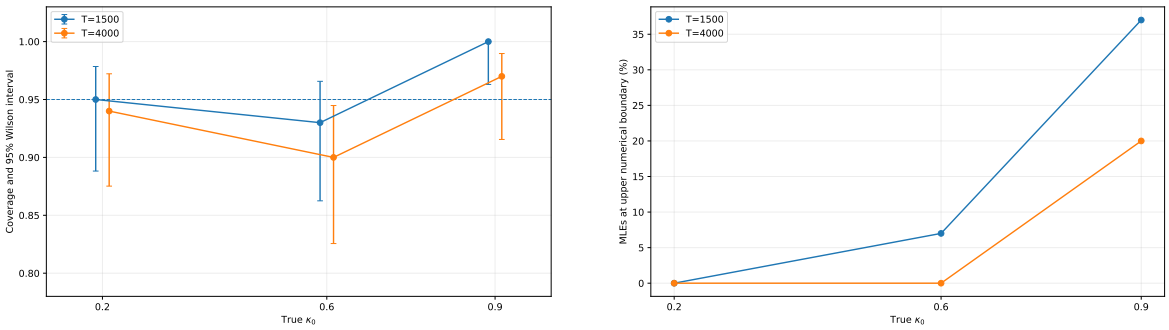


Figure 3: Experiment 15, primary 100-replication cells. Left: coverage with 95% Wilson intervals for estimated coverage probability and nominal 0.95 reference; right: frequencies of fitted upper-bound κ. Boundary counts refer to MLEs, not the true parameter lying on a mathematical boundary.

upper-bound frequencies. The 90% coverage at $(\kappa_0, T) = (0.6, 4000)$ warrants follow-up, but is a cell among six examined. The 100% coverage at $(0.9, 1500)$ is consistent with a conservative finite-sample outcome associated with frequent constrained estimates, but does not establish a causal mechanism. In particular $\kappa_0 = 0.9$ is still an interior value of $[0, 0.999]$; boundary-asymptotic mixture distributions cannot simply be asserted for this data-generating point. The numerical upper box nonetheless strongly influences the finite-sample fit.

Crucially, profile *width* and bootstrap *calibration* were computed only for deliberately selected subgroups. The 90 interval-shape observations consist of 15 per cell: for $\kappa_0 = 0.9$, the upper endpoint hit 0.999 in 15/15 small-sample intervals and 14/15 large-sample intervals. The null-bootstrap study selected just *six outer samples per cell*, with 59 bootstrap fits for each. The 36 resulting coverage comparisons and 2,124 intended inner simulations cannot establish superior calibration. They are not a full inversion of bootstrap-calibrated confidence sets.

6.5 Analytic explanation: an exact kernel metric and a weak direction

Let $x_k = k - 1 + \delta$, $X_k = \log x_k$, $Y_k = x_k^{-1}$ and $\mathbb{E}_w g = \sum_k w_k g_k$. The log-weight derivatives are

$$\partial_\alpha \log w_k = -(X_k - \mathbb{E}_w X), \qquad \partial_\delta \log w_k = -\alpha(Y_k - \mathbb{E}_w Y). \tag{4}$$

Treating the normalised weights as a discrete probability vector defines the positive-semidefinite score Gram matrix

$$G_{ab}(\alpha, \delta) = \sum_{k=1}^{K} w_k \, \partial_a \log w_k \, \partial_b \log w_k = \begin{pmatrix} \mathrm{Var}_w(X) & \alpha \, \mathrm{Cov}_w(X, Y) \\ \alpha \, \mathrm{Cov}_w(X, Y) & \alpha^2 \, \mathrm{Var}_w(Y) \end{pmatrix}. \tag{5}$$

Its determinant is $\alpha^2 \, \mathrm{Var}_w(X) \, \mathrm{Var}_w(Y)(1 - \mathrm{Corr}_w(X, Y)^2)$. For $K \geq 3$ and strictly positive parameters this is *strictly positive*, so the exact kernel map remains locally identifiable, yet a correlation near -1 produces a small eigenvalue and an elongated local ridge. A Taylor expansion about a representative lag x_* gives $\log x \simeq \log x_* + (x - x_*)/x_*$ and $x^{-1} \simeq x_*^{-1} - (x - x_*)/x_*^2$: the two score vectors are approximately negatives of one another up to a scale factor. In the $\delta \gg K$ limit,

$$\log \frac{w_k}{w_{k+1}} = \alpha \log\left(1 + \frac{1}{k - 1 + \delta}\right) \simeq \frac{\alpha}{\delta}, \tag{6}$$

so the data primarily see the ratio α/δ, and moving the two parameters together may leave the weights nearly unchanged. This is approximate, not exact, invariance.

At $(K, \alpha, \delta) = (250, 1.8, 5)$, an independent deterministic evaluation of (5) gives

$$G \approx \begin{pmatrix} 0.858311 & -0.096712 \\ -0.096712 & 0.013004 \end{pmatrix}, \quad \mathrm{Corr}_w(X, Y) \approx -0.915406,$$

and a log-parameter-coordinate metric condition number of approximately 63.84. The last number is coordinate-dependent; it is *not* the condition number of the observed statistical information matrix. Figure 4 illustrates the implied near-equivalent kernel curves without passing them off as fitted likelihood profiles.

To connect this kernel geometry to return-likelihood information, hold observed u_{t-k} fixed and write, for $\vartheta = (\alpha, \delta)$,

$$\partial_{\vartheta_a} \log h_t = \frac{\rho}{h_t} \sum_{k=1}^{K} u_{t-k} \, \partial_{\vartheta_a} w_k. \tag{7}$$

For *known* Student-t degrees of freedom ν, let S_t be the scale log-likelihood score for $\log h_t$. It has conditional mean zero and

$$\mathbb{E}(S_t^2 \mid \mathcal{F}_{t-1}) = \frac{\nu}{2(\nu + 3)}. \tag{8}$$

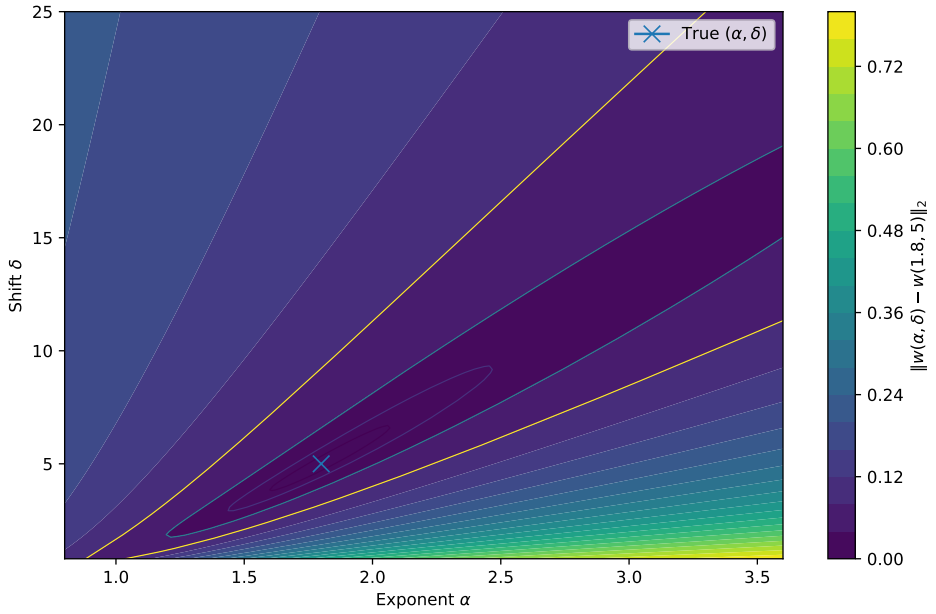


Figure 4: *Analytical kernel-distance map*, not an observed likelihood: Euclidean separation of $w(\alpha, \delta)$ from $w(1.8, 5)$, $K = 250$. The valley is a geometric explanation for, but not an independent reproduction of, the likelihood ridge reported in Experiment 13.

Thus the information for the shape coordinates, before profiling nuisance parameters, has the form

$$\mathcal{I}_{\vartheta\vartheta} = \frac{\nu}{2(\nu+3)} \sum_t \mathbb{E}\left[(\nabla_\vartheta \log h_t)(\nabla_\vartheta \log h_t)^\top\right]. \tag{9}$$

It is *not equal* to the simple kernel metric G: it further depends on the stochastic history and on the simultaneous estimation of $\omega, \rho, \kappa, \nu$. Profiling nuisance parameters subtracts a positive-semidefinite information term, potentially flattening an already shallow direction. The kernel result therefore motivates, but does not by itself prove, weakness of statistical identification.

6.6 Wilks' limit, numerical boundaries and reporting implications

Classical likelihood-ratio theory yields a χ^2_1 limit under regular interior identifiability, suitable mixing/moment conditions and consistently solved likelihood optimisation (Wilks 1938; Robinson and Zaffaroni 2006). These asymptotic conditions do not guarantee accurate approximation for a fixed T. Flat nuisance ridges, $T - K$ effective contributions, finite numerical bounds, and heavy-tailed volatility dynamics can all affect practical calibration. For true parameters *on* model boundaries, non-standard LR laws may arise (Self and Liang 1987; Andrews 2001). A large fraction of *estimated* coefficients at numerical bounds does not itself determine the asymptotic null distribution for an interior generating coefficient.

The studies motivate three practical principles: report joint (α, δ) profiles rather than only standard errors; accompany every profile coverage estimate with interval width, connectedness, bound hits and optimiser diagnostics; and assess forecast uncertainty separately from parameter uncertainty. For the latter, with fixed (ω, ρ, κ) and a common observed history, the bound

$$|\widehat{h}_t - h_t| \le \rho \|\widehat{w} - w\|_1 \max_{1 \le k \le K} u_{t-k} \tag{10}$$

is immediate. It clarifies why similar kernels can deliver similar forecasts even when their (α, δ)

coordinates differ sharply. The right-hand side is path-dependent and is not a guarantee of forecasting accuracy when persistence, intercept or asymmetry estimates also differ.

7 Experiments 16–17: misspecification, forecast functionals and predictive inference

7.1 Experiment 16: forecast functionals and deliberate misspecification

7.1.1 Synthetic design and reported-information boundaries

Experiment 16 studies forecast-functional uncertainty and model misspecification beyond the correctly specified Monte Carlo settings of Experiments 13–15. Its four data-generating processes (DGPs) are symmetric DARM, asymmetric DARM, Student-t GJR-GARCH, and latent log-stochastic volatility (log-SV); each has the same *specified* unconditional variance target $0.02^2 = 0.0004$. At each $T \in \{1500, 4000\}$, 30 independently seeded paths per DGP give 240 paths in total. The supplied design declares 1,600 burn-in and a 500-day holdout; DARM uses $K = 250$, and the stated generating innovations have $\nu = 8$ except that the full log-SV noise mechanism is defined only by design parameters and not recoverable from its missing executable notebook. Five forecasting methods are compared: symmetric and asymmetric DARM, a restricted $\gamma \geq 0$ Student-t GJR-GARCH with subcritical persistence enforced by a coefficient transformation, fixed $\lambda = 0.94$ EWMA and a 252-day rolling variance estimator. The synthetic GJR estimator is *not identical* to the earlier empirical test's `arch` implementation. For each successful replication the archived score file contains 500 one-day daily losses, 99 five-day weekly losses and 99 ten-day weekly losses per model. Ten days is the principal *synthetic diagnostic*, not a new market-data confirmatory trial.

The audit was performed on the eight included CSV files, `design.json`, the supplied diagnostic figures and handoff documentation. *The handoff ZIP does not contain the Experiment 16 source notebook, seed-specific realised returns and latent states, per-origin predictions, or the described NPZ/checkpoint files.* Accordingly the calculations below reconstruct replicate-level scores, pairwise differences, Student-t confidence intervals and functional errors, but cannot independently establish temporal filtration correctness, oracle formulas, simulated input paths or optimisation convergence beyond the 240 success flags. Forecast origin counts are consistent with a 500-day holdout and weekly origins but do not by themselves rule out future-data leakage. The nominal $\sigma = 2\%$, initialisation and Oracle conditionings are taken from the archived design, rather than verified from execution traces.

7.1.2 Two distinct risk summaries: expected oracle regret and realised paired loss

Let $V^{\star}_{t,H}$ denote the DGP-specific oracle conditional integrated variance under its stated information set and $\widehat{V}_{t,H}$ a fitted forecast. For the standard variance QLIKE score $L(y, v) = \log v + y/v$, the *oracle-expected excess loss* at a forecast origin has the form

$$D(\widehat{V}, V^{\star}) = \mathbb{E}_{\star}[L(RV, \widehat{V}) - L(RV, V^{\star})] = \log(\widehat{V}/V^{\star}) + V^{\star}/\widehat{V} - 1 \geq 0, \tag{11}$$

provided $V^{\star} = \mathbb{E}_{\star}(RV)$ under the oracle conditional law, and the comparison takes both forecasts at the same origin. The archive labels its first summary *expected QLIKE regret versus the DGP oracle*; its exact computational formula cannot be independently read from the absent notebook. By contrast, the *paired realised* difference for candidate A versus benchmark B on replication r is

$$d_r(A, B) = \overline{L(RV, \widehat{V}^{B})}_r - \overline{L(RV, \widehat{V}^{A})}_r, \tag{12}$$

where the bar averages forecast origins *within* the replicated path. Its inference uses $n = 30$ independent *paths*, not the 99 overlapping weekly forecasts as independent units. Positive d_r favours

A. These quantities are not interchangeable: their pointwise targets are different, and realised proxies add conditional noise. The log-SV oracle observes latent-state information unavailable to the fitted return-only methods, so its oracle regret provides an *informationally privileged* reference, not an implementable model comparison. The distinction is grounded in proper conditional scoring and volatility-proxy theory (Gneiting and Raftery 2007; Patton 2011); latent state modelling is well established (Kim et al. 1998).

Table 4: Experiment 16: ten-day *expected oracle-relative QLIKE regret*, reported means over 30 independent replications per cell. Lower is better. These values are **not** paired realised QLIKE gains. Model order is symmetric DARM (S), asymmetric DARM (A), GJR (G), EWMA (E), rolling variance (R).

DGP	T	S	A	G	E	R
Symmetric DARM	1500	**0.01003**	0.01161	0.01644	0.03680	0.06576
Symmetric DARM	4000	**0.00298**	0.00302	0.00640	0.03913	0.05640
Asymmetric DARM	1500	0.01693	**0.01008**	0.01712	0.04750	0.08308
Asymmetric DARM	4000	0.00915	**0.00221**	0.00839	0.04961	0.07696
GJR-GARCH	1500	0.00776	**0.00429**	0.00439	0.05267	0.06080
GJR-GARCH	4000	0.00656	0.00278	**0.00235**	0.05803	0.07156
Log-SV	1500	0.10198	0.10224	**0.10144**	0.12360	0.24576
Log-SV	4000	**0.09542**	0.09572	0.09561	0.13541	0.22688

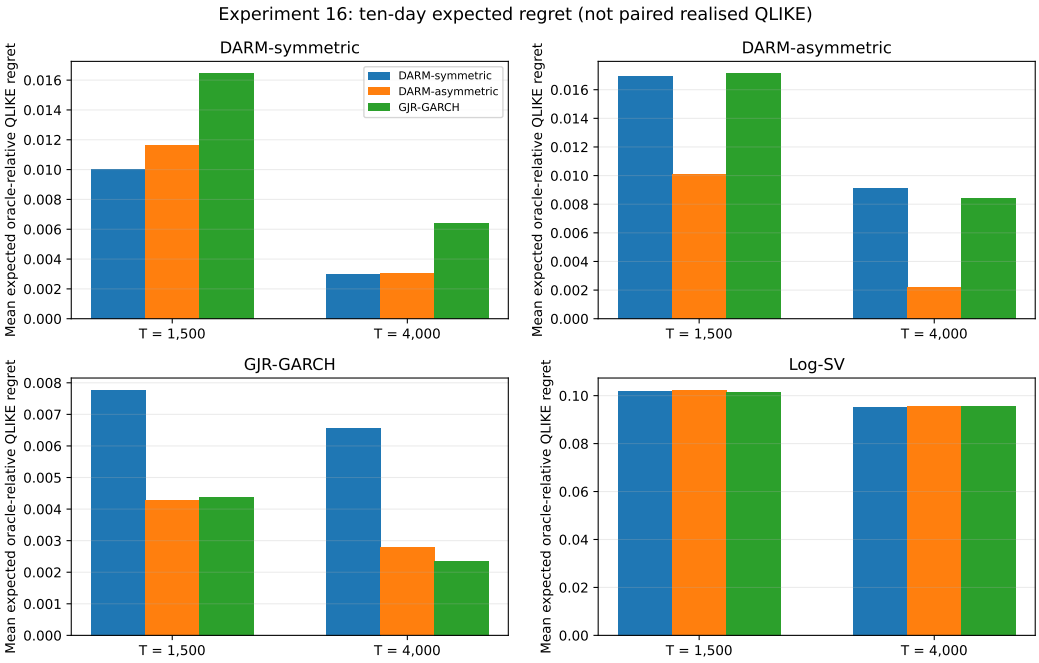


Figure 5: Reported mean ten-day oracle-expected QLIKE regret under all four synthetic DGPs; plotted from replicate-summary CSV. These are not realised paired-loss contrasts or tests of model dominance.

7.1.3 Paired ten-day comparisons: no general superiority claim

The $T = 1500$ symmetric- and asymmetric-DARM DGP cells have unadjusted intervals above zero; under the GJR-GARCH and log-SV DGPs all four relevant DGP-cell contrasts are compatible with zero. In particular, under log-SV fitted DARM and GJR are close in their realised paired losses, yet far from the more informative latent-state oracle in expected-regret terms. Under GJR-generated data at $T = 4000$, the oracle-regret criterion ranks GJR marginally ahead of asymmetric DARM (0.00235 versus 0.00278), while the realised paired difference slightly favours asymmetric DARM ($+0.001024$). These are different estimands and different noisy summaries, not an arithmetic contradiction. At

Table 5: Experiment 16: *paired realised ten-day QLIKE improvement* of asymmetric DARM over GJR-GARCH, averaged across independent replications. Positive values favour DARM. The 95% intervals are unadjusted Student-t intervals, $n = 30$ and $df = 29$.

DGP	T	Mean $\bar{d}$	95% interval
Symmetric DARM	1500	$+0.004036$	$[+0.000165, +0.007908]$
Symmetric DARM	4000	$+0.003358$	$[-0.000237, +0.006952]$
Asymmetric DARM	1500	$+0.007990$	$[+0.002138, +0.013843]$
Asymmetric DARM	4000	$+0.006183$	$[-0.002901, +0.015267]$
GJR-GARCH	1500	$+0.000470$	$[-0.001034, +0.001974]$
GJR-GARCH	4000	$+0.001024$	$[-0.000609, +0.002657]$
Log-SV	1500	-0.000746	$[-0.004102, +0.002610]$
Log-SV	4000	-0.000499	$[-0.002424, +0.001425]$

30 replicates per cell, multiple testing across DGPs, horizons and candidate pairs limits claims of statistical significance. The archive reports 144 comparison cells in total; unadjusted intervals should not be read as a familywise decision rule.

7.1.4 Inference on memory functionals and asymptotic half-life

For a finite strictly positive kernel write $W(R) = \sum_{k=1}^{K} w_k R^k$ and let $R > 1$ be the unique positive solution of $\rho W(R) = 1$. The asymptotic renewal-response half-life is $H_{1/2} = \log 2 / \log R$. It is a decay *functional* of the linear renewal response, not necessarily the half-life of every realised nonlinear volatility shock. From implicit differentiation,

$$\frac{\partial H_{1/2}}{\partial \rho} = \frac{\log 2}{\rho^2 R W'(R) (\log R)^2} > 0, \qquad \frac{\partial H_{1/2}}{\partial \theta} = \frac{\log 2 \, \partial_\theta W(R)}{R W'(R) (\log R)^2}, \tag{13}$$

for each kernel parameter θ. For the normalised power-law kernel, $\partial_\alpha W(R) = -\operatorname{Cov}_w(R^k, \log(k - 1 + \delta)) < 0$ and $\partial_\delta W(R) = -\alpha \operatorname{Cov}_w(R^k, (k - 1 + \delta)^{-1}) > 0$ for nondegenerate $K \geq 2$. Thus half-life decreases with the power exponent α and increases with shift δ (holding ρ fixed). This is a deduction for the chosen parameterisation, not a new stochastic-process existence result. As $\rho \uparrow 1$, when the finite kernel is held fixed with mean lag $\bar{k}$, $R - 1 \sim (1 - \rho)/(\rho \bar{k})$ and

$$H_{1/2} \sim \frac{(\log 2) \rho \bar{k}}{1 - \rho}. \tag{14}$$

Consequently even small estimation errors in ρ can create large half-life errors near criticality. The exact root calculation for $K = 250, \alpha = 1.8, \delta = 5, \rho = 0.75$ yields $R = 1.0099825002$ and $H_{1/2} = 69.78223$ days, matching the functional truth recorded in the archive. This calculation is independent of fitted stochastic paths.

Table 6: Experiment 16: response half-life error for the correctly matched DARM estimator under its generating DGP. Each cell has 30 replications; these are *sampling errors of point estimators*, not interval coverage estimates. True half-life 69.78223 days in all cells.

True DGP	T	Bias (days)	RMSE (days)	Median absolute error
Symmetric DARM	1500	-17.86	39.58	31.13
Symmetric DARM	4000	-18.63	29.73	25.86
Asymmetric DARM	1500	-4.42	44.70	40.54
Asymmetric DARM	4000	-3.02	39.85	24.60

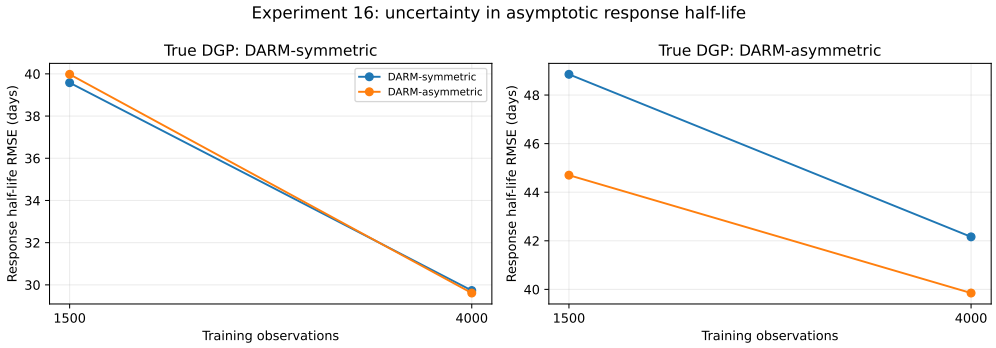


Figure 6: Monte Carlo RMSE of the finite-K dominant-pole response half-life across 30 independent synthetic datasets per cell, for correctly and incorrectly specified DARM fits to DARM-generated data.

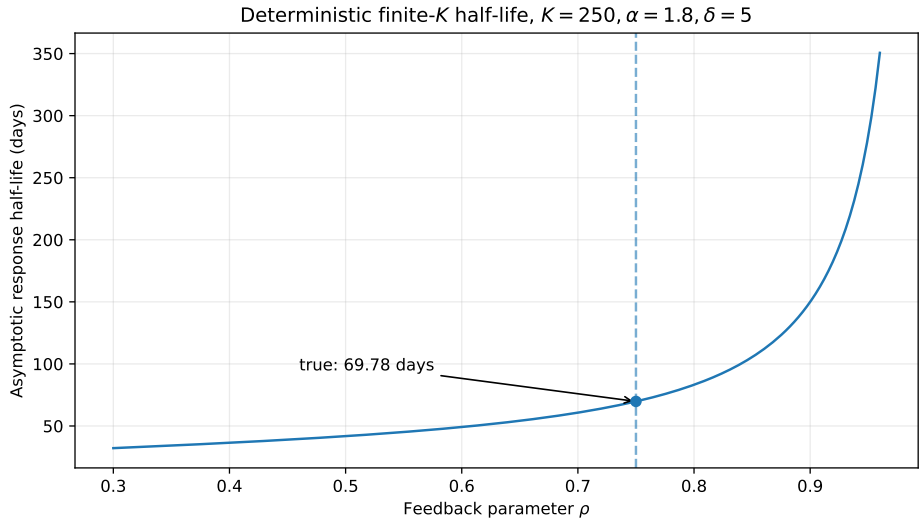


Figure 7: Analytical sensitivity of the finite-lag half-life to ρ at the generating kernel; purely deterministic calculation, not a Monte Carlo forecast or empirical market result.

The true mean memory lag is 19.32971 and the true effective lag count is 17.07654 for the chosen power-law kernel. Kernel coordinates may be practically weakly identified even when short-horizon conditional variances are forecast accurately, because short-horizon scores respond to a finite set of recent weighted shocks whereas the pole and its inferred half-life depend on the entire kernel and the feedback parameter through an implicit nonlinear equation. *Inferred half-life uncertainty should therefore be reported separately from forecast accuracy*, preferably through joint parameter-profile or bootstrap intervals with evaluated calibration. The archive's 36 exploratory percentile intervals cover six quantities for three *selected* outer replications in each of two $T = 4000$ DARM cells, with just $B = 39$ inner bootstrap fits. Selection prevents using their observed containment proportions to estimate unconditional coverage, and the coarse Monte Carlo quantiles preclude fine calibration claims.

7.1.5 Limitations specific to the new synthetic study

The score summaries are internally consistent to machine precision: 3,600 model-horizon-replication scores imply 120 model-performance rows, and 4,320 paired replicate differences imply 144 contrast summaries. Their reported intervals are exactly reconstructed with the 29-degree-of-freedom Student-t critical value. These checks do *not* replicate the innovations, oracle targets, forecast chronology or likelihood maximisation. The stated zero failures are zero *recorded failures*, not proof of globally optimal fits. The GJR comparator uses positive-γ restrictions; the latent log-SV model's full unobserved state is privileged in the oracle benchmark. No confidence interval corrects for all 144 comparisons. The nominal common unconditional variance and matched targets need the omitted notebook/raw paths to verify independently. The full-run results must be distinguished from any preliminary pilot runs and from selected bootstrap diagnostics. None of these simulated outcomes changes the 2020–2025 market-data primary ten-day non-significance result in the main paper's financial-evidence section.

7.2 Experiment 17: return-filtered SV and expanded misspecification benchmarks

7.2.1 Design, information sets and verification boundary

Experiment 17 reports $4 \times 2 \times 30 = 240$ independent simulated datasets, with symmetric DARM, asymmetric DARM, Student-t GJR-GARCH and latent log-stochastic volatility as generating processes, $T \in \{1500, 4000\}$, a declared burn-in of 1,600 periods, a 500-observation forecast evaluation sample, and $\nu = 8$ for innovations. The seven candidate models are symmetric and asymmetric power-law DARM, GJR-GARCH, filtered stochastic volatility (SV), HAR-squared, EWMA ($\lambda = 0.94$), and rolling variance (252 periods). Each replicate contains 99 evaluated origins at a five-observation step for *every* horizon $H \in \{1, 5, 10\}$, so the one-day Experiment 17 score is not directly comparable with the 500 daily one-day origins of Experiment 16. The reported principal synthetic horizon is ten days. Successful run flags cover all 240 datasets; five fitted-model parameter records appear per dataset, the two remaining procedures being parameter-light baselines.

The fitted filtered-SV procedure uses training-only Gaussian QML applied to transformed log-squared returns for the latent AR(1) parameters and subsequently filters observed returns with a Student-t measurement likelihood (Harvey et al. 1994; Kim et al. 1998; Durbin and Koopman 1997). The fitted procedure *fixes* Student-t degrees of freedom at the true $\nu = 8$, while the fitted DARM and GJR models estimate ν. Thus its training-information advantage is not eliminated by return filtering. The generating-parameter oracle for log-SV uses a stated 61-state discretised Bayesian filter conditioned on *observed returns through the origin*, not the true latent path. This improves the oracle's information-set comparability relative to Experiment 16, whose latent-state oracle was privileged. Nevertheless the generating-parameter oracle remains infeasible for a genuine unknown-DGP forecasting problem;

discretisation error is a separate unverified issue. The benchmark named HAR-squared regresses *daily squared returns* using 1-, 5- and 22-day averages and positive least squares, and must not be described as HAR based on observed high-frequency realised variance (Corsi 2009; Andersen et al. 2003).

The handoff archive contains eight CSV files, two PNG plots, a design specification and README. Crucially, it contains *no executable Experiment 17 simulation/filtering source*, per-origin model predictions, realised return paths, oracle conditional means, optimiser convergence traces or promised replicate-level NPZ/checkpoint paths. Accordingly the documented chronology is conceptually nonanticipative, and 99 five-step origins are consistent with the design, but absence of future-data leakage, correct oracle conditioning, grid filtering and horizon alignment cannot be confirmed by source-code inspection. We independently reconstruct the aggregate arithmetic from the full replicate-level *score* records; that is a narrower, but reproducible, verification claim.

7.2.2 Expected oracle-relative loss is not paired realised loss

Let $\mathcal{I}_t$ be observed information through forecast origin t and $V^\star_{t,H} = \mathbb{E}[RV_{t,H} \mid \mathcal{I}_t, \theta_0]$ the oracle conditional expected integrated variance, with known generating parameters θ_0. For a positive forecast v, conditional QLIKE has its minimum at $V^\star$, and expected regret equals

$$\mathcal{R}_t(v, V^\star) = \log \frac{v}{V^\star} + \frac{V^\star}{v} - 1 \geq 0, \quad \mathcal{R}_t = \frac{(v - V^\star)^2}{2(V^\star)^2} + O\left(\left|\frac{v - V^\star}{V^\star}\right|^3\right). \tag{15}$$

The inequality follows from $\log x + x^{-1} - 1 \geq 0$, with equality precisely at $x = 1$. The quadratic approximation is local and does not equate parameter estimation accuracy with prediction accuracy. The archived `mean_expected_regret_vs_return_info_oracle` aggregates this quantity *as reported*; the archive does not contain per-origin $V^\star$ values needed to recalculate the underlying regret without relying on generated summaries.

For a candidate C and benchmark B, the *realised* paired contrast is, separately,

$$d_r(C, B) = \frac{1}{99} \sum_{t \in \mathcal{O}_r} \Big[\mathrm{QLIKE}(RV_{t,H}, \widehat{V}^B_{t,H}) - \mathrm{QLIKE}(RV_{t,H}, \widehat{V}^C_{t,H})\Big]. \tag{16}$$

There are 30 independent dataset-level values d_r per specified DGP/T/H cell; the reported Student-t interval has centre $\bar{d}$ and half-width $t_{.975,29} s_d/\sqrt{30}$. Overlapping 5- and 10-day windows within a dataset are *not* treated as independent repetitions. Conditional expectation of realised paired loss differences corresponds to expected loss differences at matching origins under suitable oracle assumptions, but finite-sample realised differences are not arithmetically identical to differences between tabulated expected-oracle-regret means. The latter and the former must not be interchanged (Patton 2011; Diebold and Mariano 1995).

7.2.3 Ten-day expected-regret performance

The correctly specified DARM estimator achieves the lowest regret under each DARM DGP, with a slight symmetric-model advantage over asymmetric DARM in the symmetric data. Under a GJR generator, asymmetric DARM and GJR are close at $T = 1500$, while GJR has the lower reported expected regret at $T = 4000$. Under log-SV, filtered SV reduces ten-day regret from 0.02131 (GJR) and 0.02201 (asymmetric DARM) to 0.00896 at $T = 1500$, and from 0.01074 and 0.01097 to 0.00279 at $T = 4000$. This is a synthetic DGP-specific finding, not proof of a universally dominant filtering architecture. Comparison of Experiment 16 and 17 oracle-relative *levels* under log-SV is not a controlled model-only ablation: the oracle information set changes from true-state access to return-filtered known-parameter conditioning, and full equality of realised data paths and numerical procedures cannot be checked from the handoffs.

Table 7: Experiment 17: mean ten-day *expected QLIKE regret relative to the generating-parameter return-information oracle*, from 30 independent datasets per DGP/training-length cell. Lower is better. Column order: symmetric DARM (S), asymmetric DARM (A), GJR (G), return-filtered SV (F), squared-return HAR (H), EWMA (E), rolling 252-day (R). These are not realised matched-path comparison statistics.

DGP	T	S	A	G	F	H	E	R
Symmetric DARM	1500	**.00999**	.01102	.01576	.02425	.01753	.03639	.06578
Symmetric DARM	4000	**.00316**	.00321	.00662	.01079	.00887	.03817	.05499
Asymmetric DARM	1500	.01715	**.01048**	.01844	.02959	.03288	.04861	.08089
Asymmetric DARM	4000	.00887	**.00225**	.00848	.02232	.01816	.04924	.07724
GJR-GARCH	1500	.00764	**.00427**	.00438	.01635	.01202	.05368	.06043
GJR-GARCH	4000	.00666	.00278	**.00234**	.01428	.00991	.06036	.07269
Log-SV	1500	.02178	.02201	.02131	**.00896**	.03699	.04465	.16207
Log-SV	4000	.01085	.01097	.01074	**.00279**	.02715	.04687	.15651

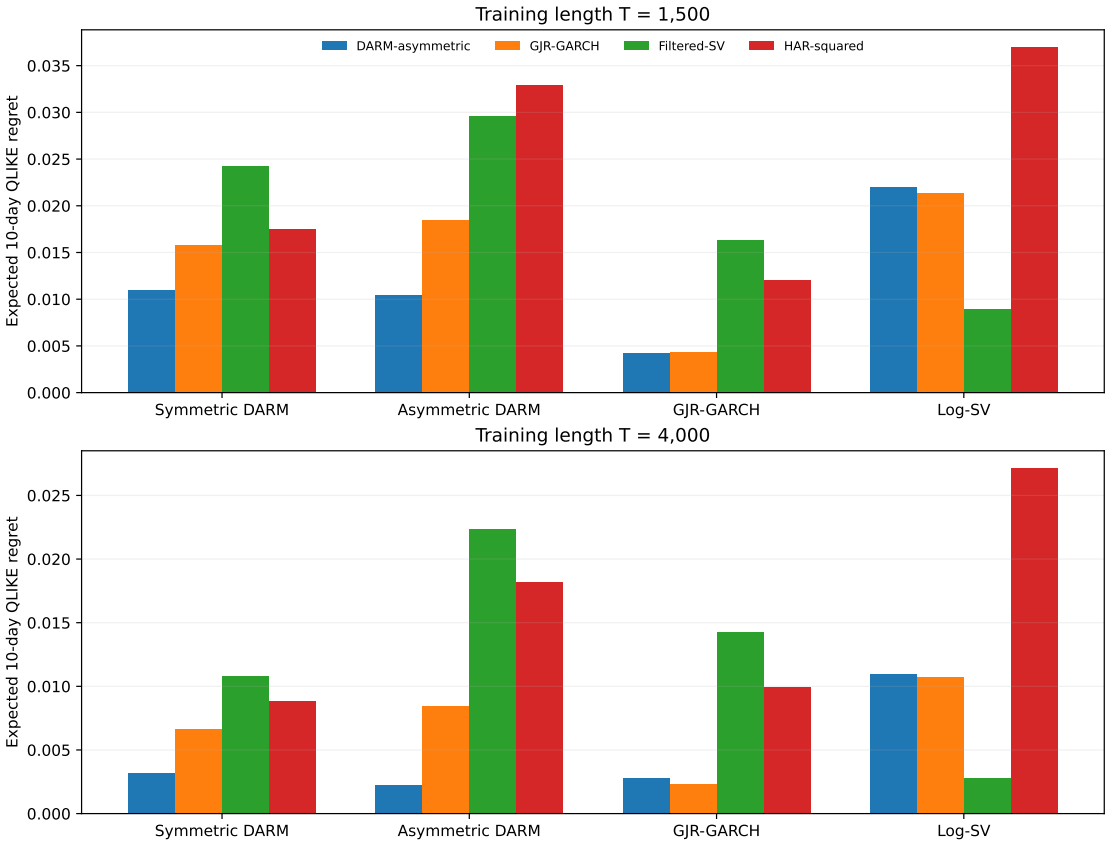


Figure 8: Expected ten-day generating-process oracle-relative QLIKE regret for four selected fitted models. The two training lengths are displayed separately. A difference between regret means is *not* the realised paired-path comparison in Table 8. Values are reconstructed from the archived summary columns.

7.2.4 Paired realised scores and statistical interpretation

Table 8: Experiment 17: ten-day *realised paired QLIKE improvements* across $n = 30$ independent datasets per cell. Positive values favour the candidate named in the contrast. The 95% Student-t intervals are *unadjusted* for numerous model/horizon/DGP contrasts.

Generating DGP and candidate–benchmark	T	Mean gain	95% interval
Asym DARM: A–GJR	1500	+.009652	[+.003559,+.015746]
Asym DARM: A–GJR	4000	+.006204	[-.002878,+.015287]
GJR: A–GJR	1500	+.000514	[-.000988,+.002016]
GJR: A–GJR	4000	+.001207	[-.000438,+.002851]
Log-SV: A–GJR	1500	-.000678	[-.004041,+.002685]
Log-SV: A–GJR	4000	-.000184	[-.002057,+.001689]
Log-SV: F–GJR	1500	+.004011	[-.005764,+.013787]
Log-SV: F–GJR	4000	+.010951	[+.004755,+.017147]
Log-SV: F–A	1500	+.004689	[-.005619,+.014997]
Log-SV: F–A	4000	+.011135	[+.005365,+.016905]

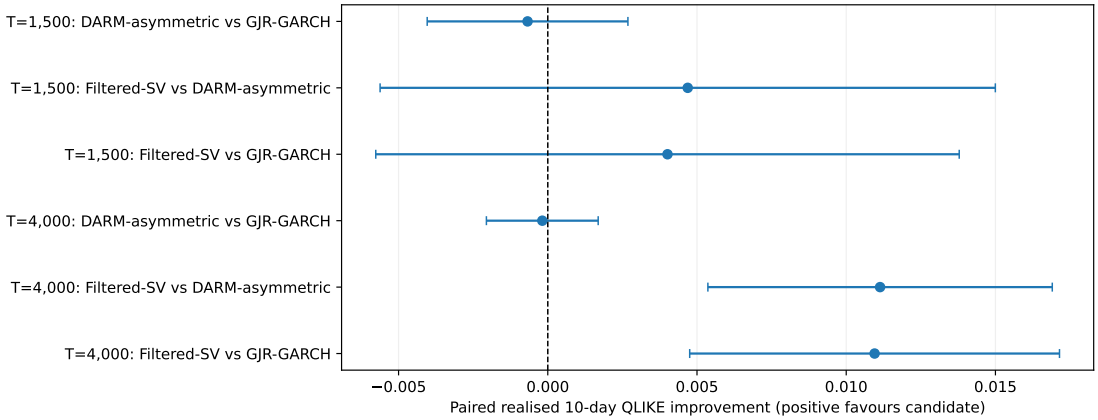


Figure 9: Log-SV-generating datasets only: 30-path paired realised ten-day QLIKE comparisons, with unadjusted 95% Student-t intervals. The horizontal zero marks equal sample-average loss. A and GJR represent asymmetric DARM and GJR-GARCH.

The $T = 4000$ log-SV paired interval for filtered SV relative to GJR or asymmetric DARM lies above zero, whereas both $T = 1500$ filtered-SV intervals include zero. DARM and GJR remain close on log-SV in their paired realised scores. The positive $T = 1500$ asymmetric-DARM-DGP A–GJR contrast likewise supports an expected correct-specification advantage within that particular synthetic configuration. However, 144 reported pairwise comparisons across models, horizons, DGPs and samples create a severe multiplicity problem. The intervals are descriptive, unadjusted; their crossing or excluding zero is not a predeclared familywise-rejection rule. Model ranking also depends on both the DGP and training length. Fixed true degrees of freedom for fitted SV may contribute to its longer-sample relative performance and prevent a clean conclusion about its advantage under symmetric estimation conditions.

7.2.5 Predictive inference: distinct targets and very small selected samples

Two additional exercises in the archive have different statistical targets. *Process-predictive* 90% intervals simulate future conditional processes with *fixed estimated parameters*; their target is a future integrated squared-return outcome $RV_{t,H}$ or latent integrated conditional variance $IV_{t,H}$. They are

neither parameter-confidence intervals nor intervals for $\mathbb{E}_t[IV_{t,H}]$. Experiment 17 uses 149 predictive simulations at only three selected outer replications, evaluated at three origins per replicate, for each reported DGP/T/model/horizon cell: nine observations per summary cell. The 64 cells and 576 archived forecast-origin rows therefore do not correspond to 576 independent prediction experiments. The aggregate intervals' reported `RV_width` and `IV_width` columns are the *median*, not the mean, of per-origin interval widths (verified exactly from per-origin CSV). At least one model/horizon records missing IV bounds; missing entries must not be interpreted as failures or successes.

Separately, the *parameter-bootstrap* 95% percentile intervals target conditional expected variance at the selected observed forecast origin and involve refitting DARM on draws simulated from fitted DARM. Only two selected $T = 4000$ datasets per DARM DGP, at horizons 1 and 10, appear in the archive: eight interval rows representing *four* selected outer datasets, with 39 successful inner fits per outer dataset. All eight reported intervals happen to contain the benchmark truth, but this is not eight independent frequentist coverage evaluations and does not establish 95% calibration. The predictive and bootstrap exercises should only motivate future repeated-outer-Monte-Carlo calibration studies with substantially larger predictive draw counts.

7.3 Synthesis: identification, filter information and model robustness

Experiments 13–15 distinguish structural identification of normalised kernel weights from practical curvature of the likelihood; Experiment 16 shows that inaccurate long-run response half-life estimates can coexist with strong short-horizon variance forecasting. In Experiment 17, the local expansion in (15) highlights why accurate conditional expectation forecasts may occur for parameter vectors far apart on a shallow likelihood ridge. Experiment 17 extends model robustness comparisons beyond constrained ARCH/GARCH and fixed simple benchmarks to a latent-state competitor estimated from returns. Under the log-SV generator, the return-filtered approach with ν fixed to its true value is substantially better in expected oracle-relative loss and better in some realised paired comparisons; under DARM generators, correctly matched DARM generally leads. These results are coherent, not contradictory, and they reinforce the absence of universal DARM forecasting superiority.

For subsequent research we recommend: (i) equalising distributional nuisance-parameter treatment (ν either estimated by all comparable models or fixed by all), (ii) retaining return-only oracle conditioning in every DGP and including oracle discretisation convergence checks, (iii) archiving exact simulation/filtering code, raw paths and per-origin $\widehat{V}$, $V^\star$, RV and state estimates, (iv) preregistering a small family of paired model contrasts at the dataset level with multiplicity treatment, (v) estimating uncertainty in $H_{1/2}$ and short-horizon expected variance jointly, and (vi) assessing interval calibration over independent outer replications rather than selected examples. These are proposals, not completed experiments; none modifies the historical 2020–2025 DARM primary null result.

7.4 Experiment 18: equalised degrees-of-freedom treatment, regime variation and predictive intervals

7.4.1 Design, statistical unit and audit scope

Experiment 18 reports a fully synthetic *joint* regime sweep: stable, baseline and stress parameter settings, crossed with four generators (symmetric DARM, asymmetric DARM, Student-t GJR-GARCH and log-stochastic volatility), training lengths $T \in \{1500, 4000\}$ and 20 independently generated datasets per cell. There are $3 \times 4 \times 2 \times 20 = 480$ independent generated datasets. Fitting is repeated on the *same* datasets with either estimated Student-t degrees of freedom ν or with ν fixed at the known generating value, yielding 960 *mode evaluations*, not 960 independent datasets. Seven forecasting models and horizons $H \in \{1, 5, 10\}$ produce 20,160 replicate-level scores. Each regime changes several features of the generating process simultaneously: the Student-t tail index, feedback, kernel

parameters, GJR persistence and latent-SV persistence and volatility-of-volatility. Thus cross-regime differences cannot identify the causal effect of a single design parameter. The indicated optimizer iteration limit is 160 and the DARM lag truncation is $K = 250$.

The models include symmetric and asymmetric DARM, GJR-GARCH, return-filtered stochastic volatility, HAR-squared, EWMA and rolling-252 variance. Estimating ν for all *parametric Student-t* candidates removes the specific true-ν advantage of fitted SV in Experiment 17; it does *not* equalise estimation principles. DARM and GJR use conditional Student-t likelihood, whereas filtered SV uses Gaussian quasi-likelihood on log-squared returns when estimating ν. HAR-squared is a regression on *daily squared returns*, not an intraday HAR-RV estimator; HAR and the other distribution-free baselines do not have a Student-t nuisance parameter. The `fixed_true` setting is an oracle-parameter *sensitivity* and not implementable when the true generating ν is unknown.

The 13-file handoff supplies the design JSON, README, nine tabular output CSVs (one empty failure file), two PNG figures, fitted parameter records, replicate-level model scores and origin-level predictive-interval records. It does *not* supply executable Experiment 18 simulation/estimation/filtering source, raw generated innovations, full forecast paths or oracle-state filtering traces. We independently reproduce the aggregate score tables, matched-sample confidence intervals and predictive coverage from the provided records; checking numerical summary arithmetic is distinct from independent regeneration of the synthetic paths. In particular, causal forecast timing and return-information oracle filtering are documented but cannot be verified line by line. There are no recorded fitting failures.

7.4.2 Two distinct variance targets and two loss summaries

At observed-information origin t, define

$$RV_{t,H} = \sum_{j=1}^{H} \varepsilon_{t+j}^2, \qquad IV_{t,H} = \sum_{j=1}^{H} h_{t+j}. \tag{17}$$

For a martingale-difference return with $\varepsilon_{t+j} = \sqrt{h_{t+j}} z_{t+j}$, h_{t+j} predictable, $\mathbb{E}(z_{t+j} \mid \mathcal{F}_{t+j-1}) = 0$ and $\mathbb{E}(z_{t+j}^2 \mid \mathcal{F}_{t+j-1}) = 1$, iterated expectations give

$$\mathbb{E}_t RV_{t,H} = \mathbb{E}_t IV_{t,H}, \qquad RV_{t,H} - IV_{t,H} = \sum_{j=1}^{H} h_{t+j}(z_{t+j}^2 - 1). \tag{18}$$

The latter is a sum of martingale differences, *not* an independent measurement error relative to the whole future $IV_{t,H}$: future conditional variances can themselves depend on earlier return shocks. Equality of the conditional means does not imply equality of predictive distributions, interval widths, tails, or quantile coverage. For a fully observed ARCH model, h_{t+1} is known at time t and one-step IV is conditionally degenerate, even though $RV_{t,1}$ remains random. In latent SV, the next state is not observable from the return history, so even one-step integrated variance retains conditional uncertainty. The origin-level records accordingly contain missing one-day IV bounds for non-SV models; these missing bounds are *not* interval failures and are excluded from one-day IV coverage.

The expected QLIKE regret of forecast $v > 0$ relative to oracle conditional mean $v^\star$ is

$$\mathcal{R}(v, v^\star) = \log(v/v^\star) + v^\star/v - 1 \geq 0. \tag{19}$$

It is a conditional-expectation functional and does not use a particular future realised squared-return path. In contrast, the realised paired loss improvement for candidate A over benchmark B is $d = \mathrm{QLIKE}(RV, \hat{v}_B) - \mathrm{QLIKE}(RV, \hat{v}_A)$ computed using *the same simulated dataset*. Its empirical average and Student-t interval across independent datasets may disagree in ranking with the expected-regret means; this is not an arithmetic contradiction. The latter scores vary with the finite future shock realisations. QLIKE has established advantages with noisy conditionally unbiased variance proxies (Patton 2011), but this does not imply the correctness of the *full predictive law* for IV or RV.

7.4.3 Degrees-of-freedom sensitivity and numerical identification

The reported fixed-versus-estimated comparison is paired within each independently generated dataset. Define the fixed-true benefit as $\bar{d}_\nu = \overline{L_{\text{estimated}} - L_{\text{fixed true}}}$; positive values favour knowing ν. All 504 supplied comparisons reproduce from matched replicate-level realised QLIKE records, including two-sided Student-t intervals with 19 degrees of freedom. In the baseline log-SV setting $T = 4000, H = 10$, the observed benefits are -0.003077 for asymmetric DARM (95% CI $[-0.008233, 0.002078]$), -0.003691 for GJR ($[-0.009000, 0.001619]$), and $+0.000872$ for filtered SV ($[-0.001847, 0.003591]$). These intervals all contain zero and cannot justify a universal efficiency advantage from fixing the tail parameter.

Importantly, the *estimated* mode retains a baseline log-SV advantage in *oracle-relative expected regret* for SV: at $T = 4000, H = 10$, the means are 0.002382 for filtered SV, 0.011279 for GJR, and 0.011543 for asymmetric DARM. Under fixed-true ν the corresponding means are 0.001484, 0.013466 and 0.013401. Hence the fitted-SV expected-regret advantage in this baseline is not solely an artefact of receiving the true ν in Experiment 17. Nonetheless, estimated-ν fitted SV reaches its upper numerical bound $\nu \approx 40$ in $139/480$ fits (28.96%) when a tolerance of $\nu \geq 39.99$ is used; the shares are 62.5% (stable), 23.125% (baseline) and 1.25% (stress). The numerical upper boundary and the log-squared-return quasi-likelihood can produce practically weak identification; boundary frequency alone does not establish a formal non-identification theorem or prove that any specific fit is inconsistent.

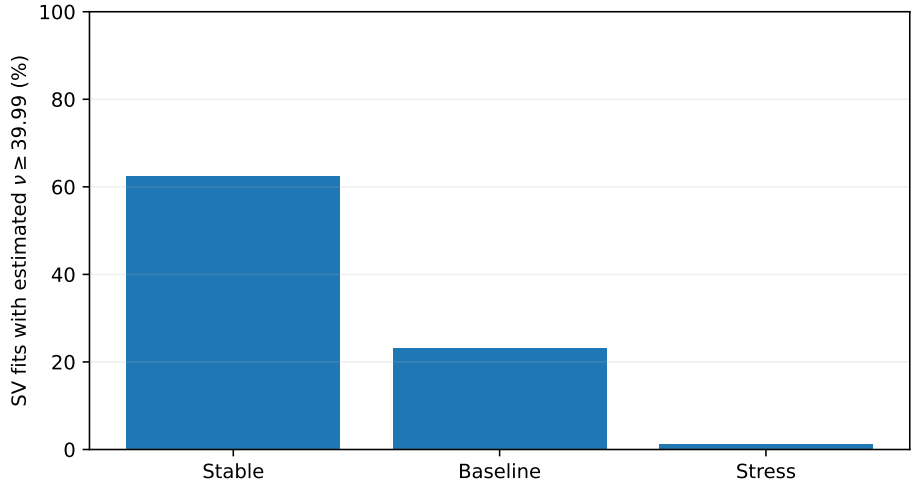


Figure 10: Experiment 18: fraction of fitted filtered-SV estimated-ν parameters near the numerical bound 40. The 480 fit records combine four generating mechanisms, two training lengths, and independent replications. Regimes jointly vary several parameters.

7.4.4 Model misspecification across regimes

Under the tested ARCH-family DGPs, DARM remains competitive with GJR; the top-ranked model varies with generator and regime. In the *stable* log-SV case, the return-filtered SV estimator does *not* dominate expected regret at $T = 4000$ (asymmetric DARM 0.000867 versus filtered SV 0.002868). In the baseline and stress log-SV cases it has appreciably lower expected regret than DARM or GJR. For the same baseline log-SV $T = 4000, H = 10$ datasets, the *realised paired* improvement for filtered SV relative to GJR is only $+0.000127$ with unadjusted 95% CI $[-0.007021, 0.007275]$, despite its lower expected oracle-relative regret. In stress log-SV, the analogous realised paired improvement is $+0.048695$ (CI $[0.014531, 0.082860]$). Thus the expected-regret ranking, realised sample contrasts and uncertainty intervals require separate reporting. Hundreds of exploratory comparisons are unadjusted

Table 9: Experiment 18, log-SV generating process, estimated-ν mode: mean ten-day *expected oracle-relative QLIKE regret*, 20 independent datasets per cell; smaller is better. Models in columns are asymmetric DARM (A), GJR (G) and return-filtered SV (F). This is *not* realised paired QLIKE.

Regime	Training T	A	G	F
Stable	1500	.001245	.001089	.004944
Stable	4000	.000867	.001023	.002868
Baseline	1500	.016488	.017662	.012782
Baseline	4000	.011543	.011279	**.002382**
Stress	1500	.065427	.062366	.020549
Stress	4000	.062992	.067145	**.008619**

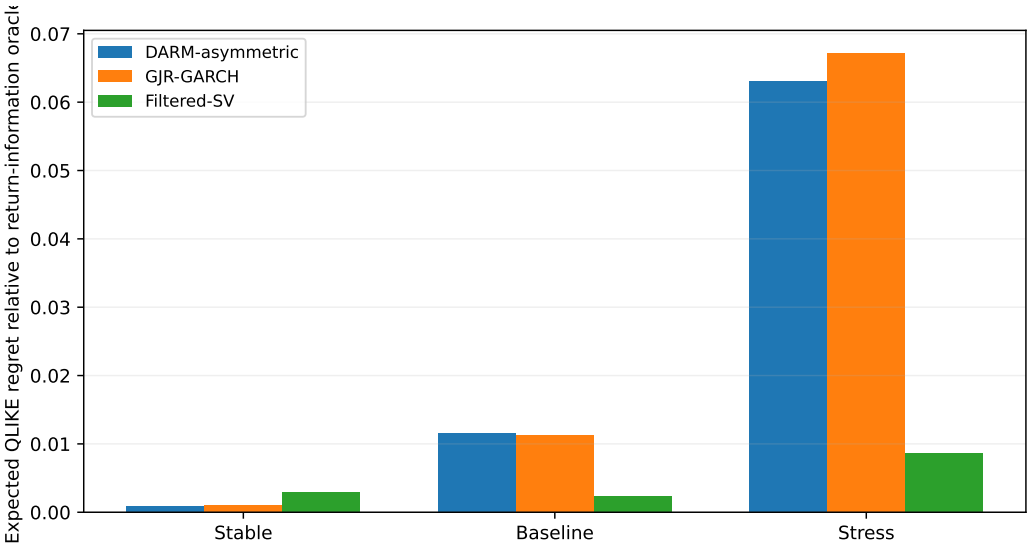


Figure 11: Ten-day expected oracle-relative regret under log-SV, $T = 4000$ and estimated ν. The stable, baseline and stress configurations are *joint* DGP changes, not one-parameter intervention contrasts.

for multiple testing; no general forecasting-superiority claim follows. The full cross-product of all DGP/regime/training-length/model/horizon combinations is supplied as CSV rather than compressing every configuration into a single paper table.

7.4.5 Ninety-percent predictive-interval calibration

The origin-level archive records 12 intervals per independent dataset/model/horizon, 199 simulated predictive draws per interval, and *estimated-ν* fits only. Reported coverage summaries average 12 within-dataset indicators and then average the 20 independent dataset-level fractions. The supplied 95% coverage-error intervals use Student-t with 19 degrees of freedom applied to those 20 independent dataset fractions; these are symmetric mean intervals, not binomial intervals with 240 independent observations. They can extend above one; 30 reported IV upper limits do, and should not be interpreted as valid probability-domain limits. Using 199 predictive draws also introduces non-negligible Monte Carlo variability into nominal 5% and 95% empirical quantiles. Parameter-estimation uncertainty is *excluded* because each predictive simulation holds fitted model parameters fixed. Interval coverage at 12 dependent selected origins per dataset does not establish uniform conditional calibration, while averaging over independently generated datasets does supply an appropriate primary inference unit.

Table 10: Experiment 18: 90% predictive-interval empirical coverage under log-SV at $T = 4000$, $H = 10$, from 20 independent datasets and 12 origins per dataset. RV and IV are different targets. Coverage is a proportion, and the intervals hold parameters fixed.

Regime	Model	RV coverage	IV coverage	median IV width
Stable	Asym. DARM	.8625	.1958	.000480
Stable	Filtered SV	.8625	.7958	.002888
Baseline	Asym. DARM	.9375	.5083	.002729
Baseline	GJR-GARCH	.9208	.4750	.002661
Baseline	Filtered SV	.9208	.9125	.005577
Stress	Asym. DARM	.8583	.4542	.001271
Stress	Filtered SV	.8833	.9083	.003099

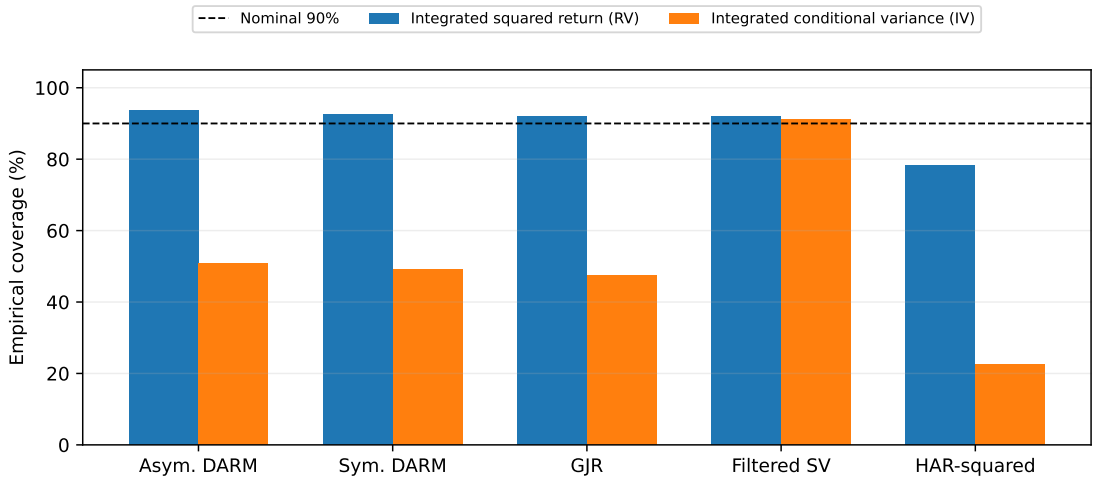


Figure 12: Baseline log-SV predictive-interval coverage of observable integrated squared returns RV and latent integrated conditional variance IV ($T = 4000$, $H = 10$). Although DARM's RV coverage is close to nominal, IV coverage is strongly deficient. Descriptive bars are not independent-origin tests.

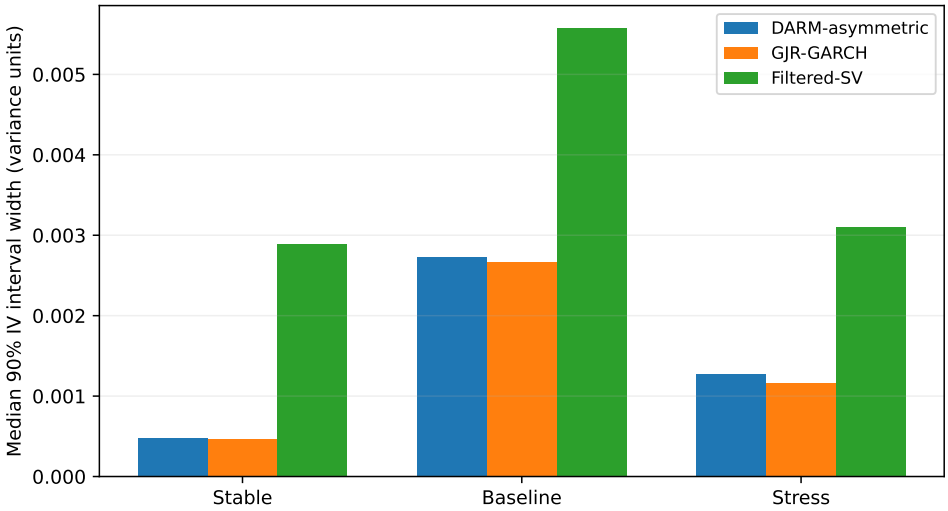


Figure 13: Median reported predictive-interval widths for latent integrated conditional variance under log-SV ($T = 4000, H = 10$), calculated directly from per-origin endpoints. Differences in both width and centring can affect empirical coverage.

For baseline log-SV $T = 4000, H = 10$, asymmetric DARM covers RV in 93.75% of origins but IV in only 50.83%, whereas filtered SV gives 92.08% RV and 91.25% IV coverage. More strikingly, asymmetric DARM's IV coverage is 19.58% in stable log-SV and 45.42% under stress. The baseline median IV interval width for asymmetric DARM is 0.002729 versus 0.005577 for filtered SV; analogous stable widths are 0.000480 versus 0.002888. Such width differences are compatible with material underrepresentation of latent-variance uncertainty, although coverage also depends on interval *location* and bias, not only width. Neither correct-looking RV coverage nor the equality of conditional means in (18) establishes a correct predictive distribution for IV. Conversely, the stronger latent-IV coverage of a model fitted to an SV DGP does not prove its superiority under ARCH-family generators or in actual equity data. These observations motivate explicit evaluation of predictive distributions for economically relevant latent risk functionals and separate treatment of parameter uncertainty (Gneiting and Raftery 2007; Christoffersen 1998).

7.4.6 Summary and reproducibility requirements

The independently recomputed 1,008 performance rows, 720 paired-comparison rows, 504 matched-mode sensitivity summaries and 360 clustered coverage rows agree with the archived summary means and intervals when the correct t_{19} quantiles and per-replicate medians are used. In particular, `median_RV_width` is a *median of within-replication medians*, not a pooled mean. Origin-level coverage records allow direct verification of the fraction covered at each origin, but do not provide the full simulated predictive distributions or the original parameter-estimation code. A full replication needs a versioned generator, fit-and-filter source, random-number streams, raw path-level returns/states, oracle conditional means, interval-draw ensembles and optimiser diagnostics. Until those are available, statements about likelihood correctness, oracle construction and causal timing remain qualified. No new independent historical-market experiment was conducted and none of the synthetic findings changes the inconclusive primary ten-day equity result.

A Full archived mathematical and empirical appendices

B Supplementary analysis of synthetic experiments

B.1 Proof of the kernel-score determinant result

For the shifted power-law kernel let $q_k = x_k^{-\alpha}$, $Z = \sum_j q_j$, and $w_k = q_k/Z$. Then

$$\partial_\alpha \log Z = -\mathbb{E}_w \log x, \qquad \partial_\delta \log Z = -\alpha \mathbb{E}_w x^{-1}.$$

This gives the centred gradients in equation (4). For three or more distinct $x_k > 0$, the functions $X \mapsto X$ and $X \mapsto \exp(-X)$ cannot be affine transforms of one another at three points because the latter is strictly convex. Hence the two nonzero centred score vectors are linearly independent in the positive weighted inner product and equation (5) is strictly positive definite. As $\alpha \to 0$ the δ score vanishes, and for $\rho \to 0$ neither kernel score influences conditional variance; both are nonregular or weakly identified limiting cases.

B.2 Information for a Student-t variance scale

Let z have standardised Student-t_ν law, with $\nu > 2$. Write $Q = z^2/(z^2 + \nu - 2)$. Under the Student-t density $Q \sim \mathrm{Beta}(\frac{1}{2}, \frac{\nu}{2})$. The score of $\log h$ for a zero-mean conditional density with observed innovation $\varepsilon = \sqrt{h}\, z$ is

$$S = -\frac{1}{2} + \frac{\nu + 1}{2} Q.$$

Since $\mathbb{E}Q = (\nu + 1)^{-1}$ and $\mathrm{Var}(Q) = 2\nu/\{(\nu + 1)^2(\nu + 3)\}$, the score has mean zero and variance $\nu/\{2(\nu + 3)\}$. The conditional information for shape parameters follows by multiplying S by $\nabla \log h_t$, which is measurable with respect to past observed returns. When ν is estimated jointly, cross-information terms must be included. The displayed information is an ideal regular reference, not proof that the actual finite-sample Hessian is stable or invertible.

B.3 Interpretation of the expected-QLIKE regret

Take $Q(h, Y) = \log h + Y/h$ and the oracle $h_0 = \mathbb{E}(Y \mid \mathcal{F}_{t-1})$, where $Y = \varepsilon_t^2$. Then

$$\mathbb{E}\{Q(\widehat{h}, Y) - Q(h_0, Y) \mid \mathcal{F}_{t-1}\} = \log(\widehat{h}/h_0) + h_0/\widehat{h} - 1.$$

This identity justifies the synthetic oracle benchmark. The inequality $\log r + 1/r - 1 \geq 0$ follows by minimisation over $r > 0$; equality is achieved only for $r = 1$. High forecast accuracy does not prove the components of a non-injective or weakly curved estimator are accurately recovered.

B.4 Archive and aggregation limitations

The CSV reports *aggregates* from executed code, so it permits consistency checks across tables and computed figures but not construction of genuine replicate-level empirical likelihood distributions. The Experiment 15 ZIP supplies four executable notebooks; it does not contain their checkpoint directories or the per-run CSV/JSON outputs. Verifying that code, input assumptions and reported aggregates are internally consistent is weaker than re-running and independently validating every fitted objective surface. Furthermore, Experiments 13 and 14 use a common RNG seed mapping and therefore must be treated as paired across matching scenario, training length and replication index. Figures in Section 6 reproduce long-form CSV aggregates or deterministic kernel arithmetic; no new historical equity price data or synthetic series were fitted during this revision.

C Experiment 16 reproducibility, forecast timing and estimands

Archived sources. The original handoff ZIP contains `design.json`, the handoff README, eight CSV result and summary files, three PNG figures, and a research-writer prompt. It does *not* contain an executable Experiment 16 notebook, per-replication NPZ return/variance paths, or optimiser checkpoint files, even though some documents refer to them as intended archive outputs. All quantitative statements in Section 7 are identified as *reconstructed arithmetic* or *reported design*, not independent code execution.

Unit of inference. For each fixed DGP, T, horizon and model pair, the 30 independent paths $r = 1, \ldots, 30$ yield paired realised-loss differences d_r. The sample standard deviation s_d uses denominator 29, and the supplied interval is precisely

$$\bar{d} \pm t_{0.975,29} \frac{s_d}{\sqrt{30}}.$$

Nested forecast origins on the same path, including 5- and 10-day targets five days apart, are not independent replicates. The stated archive contains 500 daily one-day scores and 99 weekly five-/ten-day scores per path. These counts are consistent with the documented holdout schedule, but they cannot establish nonanticipation without forecast arrays and the original code. The provided model performance table aggregates expected oracle regret, whereas the paired table aggregates realised QLIKE on matched targets; one cannot convert one to the other by subtracting the displayed row means.

Exact finite-kernel half-life derivatives. With $W(R) = \sum_k w_k R^k$ and $\rho W(R) = 1$, $W'(R) > 0$, so implicit differentiation gives $\partial_\rho R = -1/(\rho^2 W'(R))$. For a shape parameter θ, $\partial_\theta R = -\partial_\theta W(R)/W'(R)$. Since $H(R) = \log 2/\log R$, $H'(R) = -\log 2/[R(\log R)^2]$, yielding equation (13). For $\alpha > 0, \delta > 0$, the normalised weight score is $\partial_\alpha \log w_k = -\{\log(k-1+\delta) - \mathbb{E}_w \log(k-1+\delta)\}$ and $\partial_\delta \log w_k = -\alpha\{(k-1+\delta)^{-1} - \mathbb{E}_w(k-1+\delta)^{-1}\}$. The covariance identities and monotonicity follow because R^k is strictly increasing in k, $\log(k-1+\delta)$ is strictly increasing, and $(k-1+\delta)^{-1}$ is strictly decreasing. For ρ near one, Taylor expand $W(1+x) = 1 + \bar{k}x + O(x^2)$ to obtain equation (14).

Numerical checks. The companion file the provided Python verification script verifies group cardinalities, the internal consistency of all paired differences and Student-t intervals, and reproduces the full aggregate model and functional summary arithmetic. The code records maximum discrepancies at or below floating-point rounding error and checks the exact deterministic half-life root. It does not simulate or refit DARM. A full audit of forecast timing, data leakage and oracle scores requires the omitted Experiment 16 executable notebook and per-origin returns, latent states, and variance forecasts.

D Experiment 17 arithmetic reconstruction and access limitations

The Experiment 17 handoff contains eight CSV tables (5,040 score rows; 168 performance rows; 144 paired-comparison rows; 1,200 fitted-parameter rows; 240 fit-status rows; 576 predictive-origin rows; 64 predictive-summary rows; and 8 selected bootstrap rows), two PNG figures, a design JSON specification and a README. The source contains *no executable notebook, replication-level returns, oracle forecasts, or per-origin general model predictions.* Its README refers to checkpoint and NPZ path files absent from this uploaded handoff; the separately mentioned output directory is not available here.

Independent arithmetic checks group the 5,040 score rows into $4\times2\times3\times7 = 168$ unique DGP/training-length/horizon/model cells, with 30 independent datasets and 99 listed forecast origins in each. Their reported *mean* realised QLIKE, expected oracle-relative regret, realised MSE and realised/forecast ratio are reconstructed to less than 2×10^{-15} absolute difference. Pairwise matching by DGP, T, horizon and replication ID reproduces all 144 comparison means and Student-t (29 degrees of freedom) confidence limits to 2×10^{-16}. Every success flag is true. The filtered-SV fitted-parameter records have constant $\nu = 8$ for all 240 datasets. The per-origin predictive-interval rows recover each summary's coverage *and median width* exactly; mean widths do not reproduce the width columns. The eight bootstrap intervals all contain their selected truth. The supplied script `audit17.py` reconstructs the central arithmetic but not missing simulations.

These verifications establish internal consistency of *archived results*; they cannot independently certify innovation draws, unconditional-variance matching, causal forecast construction, fitted-SV QML equations, numerical filtering grids, oracle information alignment, re-estimation or the generation of interval draws. The original observed-return oracle is documented, rather than independently recomputed. The random seed is the same declared seed as Experiment 16, but absent raw paths it is also impossible to assert that the two studies used independent Monte Carlo realisations; their sample sizes must not be pooled for cross-experiment inference.

E Experiment 18 supplementary audit: record units and interval construction

The 480 independent datasets are indexed by three regimes, four generating processes, two training lengths and 20 replicate labels. The 960 fit-mode evaluations are paired within those datasets and not independent replications. For each ν treatment, seven models and three forecast horizons yield 21 scores per independent replicate. Aggregated performance has 1,008 entries. The 720 paired realised-QLIKE contrasts and 504 fixed-versus-estimated-ν contrasts are reconstructed as $\bar{d}$ from the $n = 20$ matched independent replicate differences and two-sided intervals $\bar{d} \pm t_{0.975,19} s_d/\sqrt{20}$. The comparison family is large and the intervals are exploratory without multiplicity correction.

The supplied predictive-interval file contains $86{,}400$ rows, each with 199 process simulations according to the design specification: five interval-generating models, three horizons, 12 selected origins, 480 datasets and estimated-ν mode only. The corresponding 7,200 replicate-level summaries aggregate the 12 dependent origins first; 360 coverage summaries aggregate over 20 independent datasets using $t_{0.975,19}$ times the sample standard error of their coverage fractions. Because these are symmetric intervals for a probability, some exceed unity (30 upper limits for IV). The provided RV width statistic is the median across the 20 within-replication medians, while the paper additionally calculates IV widths as the median of all available per-origin endpoint differences for selected cells. These are different summarisation operators and should be labelled accordingly.

The supplied process-predictive interval coverage indicators can be independently recomputed from [lower, upper] endpoint membership of the archived targets, subject to inclusive/exclusive equality conventions. For asymmetric ARCH, one-step latent IV is predictable given the past and has degenerate conditional distribution; its archived missing lower/upper one-step bounds are therefore not treated as uncovered observations. For log-SV, the state is latent given only observed returns, explaining why the return-filtered SV model reports one-step IV interval bounds. Full independent reproduction of the oracle, innovations, Student-t fitter, filter, and predictive draws remains impossible without source and random streams.

Archive inventory. The original handoff supplies 13 members: nine CSV outputs (one empty failure file), one JSON design, one text README and two PNG figures. The accompanying reproducibility script checks the released summaries from released replication records; it does not simulate fresh data. See the companion Experiment 18 critical audit for assumptions and caveats.

F Availability and interpretation of computational evidence

Reproducibility collection. The separate Experiments 13–18 source-and-results ZIP archive, assembled 19 October 2026 (exact filename in the main paper and release notes) includes original authored source notebooks and Python implementations for Experiments 13–17, original notebook code for Experiment 18 plus a labelled code-cell extraction, and original handoff bundles for Experiments 16–18. Executed notebook output cells exist for Experiments 13–15; a 775-row aggregate table covers Experiments 12–15 but does *not* supply original full-run per-replication fit records. The handoffs for Experiments 16–18 provide replication-level score and fitting summaries; Experiment 18 additionally archives origin-level predictive-interval records. Reconstructed aggregate summaries and audit scripts are expressly identified as *derived validation products*, not original run records.

Table 11: Evidence classes in the separate synthetic-study reproducibility archive. "Paths" means original complete simulation, filtering and optimisation trajectories, not aggregate scores.

Experiments	Source implementation	Archived outputs	Unavailable inputs
13–15	Authored Python, notebooks; executed outputs	Aggregate metrics and notebook output cells	Separate original full-run replication records and optimizer checkpoints
16	Authored Python and notebook	3,600 model-score records, fit outcomes and functionals	Complete simulated/forecast/oracle paths
17	Authored Python and notebook	5,040 model scores, fitted-parameter summaries and selected intervals	Complete return/latent-state/filter trajectories
18	Original notebook; labelled code-cell extraction	20,160 model scores, 4,800 fitted-model records and 86,400 origin interval records	Original predictive draws, latent paths and exact execution provenance

The archive's `scripts/verify_archive.py` checks content hashes and recalculates selected aggregate summaries; an audit run completed with no reported problems. This is *arithmetic validation*, not an independent end-to-end Monte Carlo reproduction. Historical code may require setting full-run mode instead of the default pilot mode, and package-version changes can alter optimisation results. The 20 independent synthetic datasets per Experiment 18 configuration, rather than the dependent forecast origins, are the replication units for coverage inference. The archive cannot establish nonanticipativity of each original filtered-oracle path solely from aggregate outputs.

Market-study limitation. The original adjusted equity-price panel, all per-origin market forecast/loss records, original full optimizer checkpoints and point-in-time constituent history are unavailable. The market study is not independently reproducible end to end from this source collection. The archive includes the historical market analysis implementation and reported summary tables as contextual material only.

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